CHE 321
Chemical Reaction Engineering
Spring 2020

INSTRUCTOR: Vikas Berry (312) 996-2342, vikasb@uic.edu

MEETING TIMES: MW 2:30 – 3:45 p.m., **ROOM:** EIB 124

TEXT: "Elements of Chemical Reaction Engineering",
Fifth, Fourth or Third Edition, by Fogler, H. Scott

OFFICE HOURS: Wednesday 3:45 – 4:45 PM (Room 226A EIB)

TA: Lan Nguyen and Seyyedfaridoddin (T: 1:30-3PM and M 10 AM-12 PM)
Tutorial: Tuesday 12-1 PM

COURSE OBJECTIVES:
This course is designed to introduce chemical engineering students with reaction engineering and to develop an expert knowledge about the various aspects of chemical reactions and kinetics. This is one of the most important courses in Chemical Engineering curriculum and will strengthen the foundation of a chemical engineering students.

Upon satisfactory completion of this course, the students will be able to:

1. Compare and contrast batch reactors, continuous stirred tank reactors (CSTR), and plug flow reactors (PFR).
2. Derive and solve the mass balance for isothermal reaction in a CSTR, PFR, or batch reactor.
3. Calculate the equilibrium conversion in any reactor type for isothermal or adiabatic conditions.
4. Set up and solve the governing mass and energy balances for a CSTR, PFR, packed bed, or batch reactor.
5. Estimate reaction orders, reaction rate constants, and Arrhenius parameters from experimental kinetic data.
6. Set up and solve the governing mass balances for multiple reactions in a CSTR, PFR, or batch reactor
7. Suggest which type of continuous reactor (CSTR or PFR) is advantageous for maximizing product selectivity for series or parallel reactions
8. Derive a Langmuir-Hinshelwood kinetic rate law
9. Incorporate a mass transfer coefficient into a reaction rate expression to account for external mass transfer limitations
10. Calculate the effectiveness factor for a catalyst particle given physical properties of the particle (pore size, tortuosity, particle diameter), the reaction rate constant, and the diffusion coefficient for the reactant
11. Suggest experiments that could be used to determine whether external and internal mass transfer is important for a heterogeneous reaction

12. Calculate the residence time distribution from a tracer study.
13. Draw similarity between electron transport through nanoparticles and chemical reactions.

GRADES:

Grades will be based on:

5 Quizzes	5 %	1 % X 5
2 Regular Exams	40 %	20 % X 2
1 Final Exam	25 %	25 % X 1
1 Project Report	12 %	12 % X 1
1 Self-Study Report	3 %	3% X 1
10 Homeworks	15 %	1.5 % X 10
	100 %	

QUIZZES: There will be a total of 7 quizzes, out of which the best five scores will go towards your grade (best 5 out of 7). **There will be no make-up quizzes.**

EXAMS: Exams will be taken during normally scheduled class periods. All the exams will be in the classroom and will consist of two sections: *Closed Book Section* and *Open Book Section*. Any necessary information relating to conversion factors and component properties will be provided for the closed book part of the exam. Test problems will be derived from lectures and homework problems. Use of only portable-hand-held calculator is permitted. The calculator must be used only for basic arithmetic calculations. No text information or graphs stored on the calculator must be used during the test. **Please bring your own paper.**

Each exam is worth 20% and each will cover half of the course. **There will be no make-up exams.** The final exam will be worth 25% grade and will be comprehensive.

If you feel the need for re-evaluation or contesting the score on an exam, you will have 10 days after you get your exam scores. After 10 days, your grade will be considered final.

PROJECT: Each student will work on a project. The project statement is provided at the end of this syllabus.

SELF-STUDY: Each student needs to do a self-study on one of the following reactors and provide a report with the following sections: (a) Introduction to the Reactor, (b) Detailed Process and its Importance, (c) Design Equations, (d) Example Solution, and (e) Safety Considerations:

- Fluidized bed Reactor
- Fermentation Reactor
- Electrochemical Reactor
- Bubble Reactor
- Spray-Tower Reactor
- Chemical Vapor Deposition Reactor
- Atomic Layer Deposition Reactor
- Hydrotreator Reactor for Diesel

HOMEWORKS: There will be a total of 10 homework assignments of 1.5 points each (best 10 out of 11).

Course Timetable

Lecture	Date	Lecture Title / Activity	Submissions/ HW/Q Dates
1	13-Jan	Course Introduction, Reaction fundamentals	
2	15-Jan	Reactor Types, General Mole Balances, Conversion	
3	22-Jan	Conversion, Reactor Sizing, Stoichiometric Tables	
4	27-Jan	Stoichiometric Tables	Q1
5	29-Jan	Isothermal Reactor Design	HW 1
6	3-Feb	SOFTWARE - Polymath and MaTLAB for Reactor Design	Q2
7	5-Feb	Pressure Drop in Reactors	HW2
8	10-Feb	Unsteady State Operation, Rate Analysis	Q3
9	12-Feb	Multiple Reactions, I	HW3
10	17-Feb	Multiple Reactions, II	
11	19-Feb	General Energy Balance & CSTR	HW 4
12	24-Feb	Review for Exam 1	
	26-Feb	EXAM 1	
13	2-Mar	PFR Energy Balance , Batch Energy Balance	
14	4-Mar	Equilibrium Conversion	HW5
15	9-Mar	Non-Isothermal Reactor Example	Q4
16	11-Mar	Multiple Steady States in a CSTR/Multiple Steady States/Multiple Reactions	
17	16-Mar	Summary of Non-Isothermal Reactors	HW6
18	18-Mar	Catalytic Reactions, Langmuir-Hinshelwood Rate Laws	Q5
		SPRING BREAK	
19	30-Mar	Langmuir-Hinshelwood Rate Laws - II	HW7
20	1-Apr	Heterogeneous Reactors, Mass Transfer	HW8
21	6-Apr	External and Internal Mass Transfer Limitations	
22	8-Apr	Effectiveness Factor and Falsified Kinetics	HW9, Q6
23	13-Apr	Review of Exam 2	
	15-Apr	EXAM 2	
24	20-Apr	TA - Residence Time Distribution II	
25	22-Apr	Zero-Parameter Flow Models	HW10
26	27-Apr	Non-Ideal Flow and combination of ideal reactors	HW11
27	29-Apr	Review of Exams	Project Due
		Final Exam	

Homework # 1

1. A CSTR reactor has to be designed for the following third-order, liquid-phase reaction:
   $3A \rightarrow 3B + C$ ($k = 0.2\ L^2/mol^2/s$)
   Stoichiometric feed of A and B enters a reactor along with an inert I. The total inlet molar flow rate of 2400 mol/min. The mole fraction of A in the inlet, $x_{AO} = 0.75$, and v_O is 100 L/min.
   (a) Make the stoichiometric table for the species.
   (b) Find the CSTR volume required to achieve a conversion of 0.8.
2. A *reversible*, gas-phase reaction ($A + B \leftrightarrow C + D$) is taking place in a PFR. The inlet concentration of A and B are 3 moles/L, $v_o = 0.5$ L/s, with no C or D in the feed. The forward reaction rate constant is 3 **mol/(m^3.(atm)^2.s)** and the equilibrium constant is 1. If the total reactor pressure is 3 atm:
   (a) What is the maximum conversion possible in this reaction?
   (b) Find an expression for conversion Vs volume of the reactor? For what volume of the reactor will the conversion be 0.42? [Note the units of the reaction constant]

Homework # 2

1. An isothermal reversible reaction ($A \leftrightarrow B$) is carried out in an aqueous solution. The reaction is first-order in both directions. The forward rate constant is 1 h^{-1} and the equilibrium constant is 1. The feed to the plant contains 100 kg-mol/m^3 of A and enters at the rate of 120 m^3/h. The reactor is a stirred tank of volume 180 m^3. The exit stream then goes to a separator, where one stream of 40 m^3/h with double the concentration of A at the exit of the reactor (C_A) is sent back to the CSTR in a continuous operation. The rest of the stream from the separator becomes the system exit (no reaction occurs in the separator).
   a. Find the system conversion.
   b. Compare this conversion with what you will get if there is no recycle.

Homework # 3

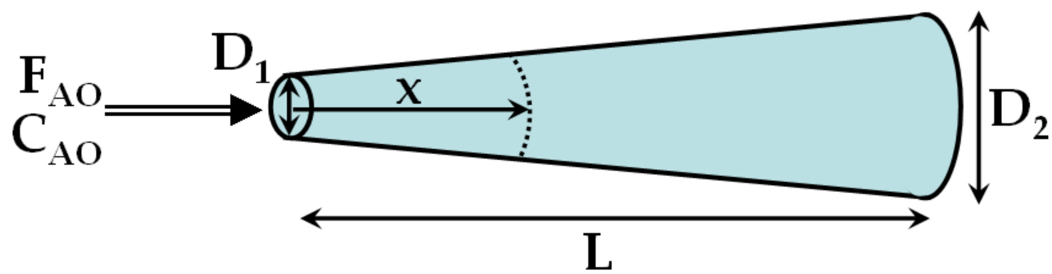
1. For a first order reaction ($A \rightarrow B$) in a PFR ($k = 1\ s^{-1}$, $\nu = 10\ dm^3/s$, $V = 100\ dm^3$), a fraction f of the exit volumetric flow rate ν is fed back to the PFR. Plot f as a function of conversion (χ).
2. (40 points) The following reactions take place in a batch reactor.
   $A \leftrightarrow D$, $\quad -r_{1A}=k_1[C_A - C_D/K_{1A}]$
   $A \leftrightarrow U$, $\quad -r_{2A}=k_2[C_A - C_U/K_{2A}]$
   $k_1 = 1.0\ min^{-1}$, $K_{1A} = 5$
   $k_2 = 100\ min^{-1}$, $K_{2A} = 2$
   $C_{A0} = 2\ mol/dm^3$

   (a) Plot and analyze conversion and the concentrations of A, D, and U as a function of time. When would you stop the reaction to maximize the concentration of D?
   (b) When does the maximum concentration of U occur?
   (c) What are the equilibrium concentrations of A, D, and U?
   (d) What would be the exit concentrations from a CSTR with a space time of 1.0 min? of 10.0 min? of 100 min?

Homework # 4

1. (40 points) A **gas phase** reaction (k_1 = 2 m^3/mol/sec at 400 K; E_A/R= 200 K) is carried out in a PFR in which the diameter changes with distance (as shown in Figure). Also a heating system is installed around the PFR such that the temperature changes along the PFR following the equation: $T = 400 + 10\ x^2$, where x is the distance (m) from the entry (left). If the entry flow rate of the reactant is F_{AO} = 1 mol/s and the entry concentration is C_{AO} = 0.5 mol/m^3, plot the conversion as a function of length of the reactor. (Assume **isobaric conditions**) [65]

$$A \rightarrow B + C\ ; \quad -r_A = K_1\ C_A^{\ 2}$$



D_1 = 0.1 m
D_2 = 0.4 m
L = 1 m
Use Polymath.

2. (20 points) What would the plot look like if the reactants enter from the right side (plot conversion versus y (y = L - x)) under same conditions?

Homework # 5

1. (100 points) A *reversible liquid-phase* reaction, 2 A⇔ B, is to be run in one or more adiabatic reactors. The rate law for this reaction is known to be:

   $-r_A = k(C_A^2 - C_B/K)$

   Where the rate constant and equilibrium constant are given by:

   $k = 600*\exp(-12000/8.314/T)$ cm^3/mol/min
   $K = 14000*\exp(\Delta H_{Rx}/R*(1/310 - 1/T))$

   where T is in K, and the enthalpy of reaction, ΔH_{Rx}, is given below.

   a) If this reaction were run in a CSTR until a temperature of 380 K, what conversion would be achieved? What volume CSTR would be required for this conversion?
   b) What is the maximum conversion you could expect to get from this reaction in a single adiabatic reactor?

c) If two large reactors (large enough to reach equilibrium) and a heat exchanger capable of returning the temperature to T_o are available, what conversion could you achieve?

<u>Useful information:</u>
C_{Ao}=0.2 mol/cm^3
C_{Bo} = 0
C_{So} = 2 mol/cm^3 (inlet concentration of the solvent)

Mean heat capacities can be used:
C_{pA} = 15 J/mol/K
C_{pS} = 20 J/mol/K (heat capacity of the solvent)

ΔH_{Rxn}(298 K)= -50,000 J/mol
ΔCp = 0
F_{Ao}=2 mol/min
T_o=310 K

Homework # 6

1. (100 points) The following reactions are taking place in an adiabatic CSTR.

$$A \xrightarrow{\Delta H_1^{\,0};k_1;E_{A1},A_1} B \xrightarrow{\Delta H_2^{\,0};k_2;E_{A2},A_2} C$$
$$B \xrightarrow{\Delta H_3^{\,0};k_3;E_{A3},A_3} D$$

[Reactions follow elementary rate law]

$A_1 = A_2 = A_3 = 1\ s^{-1}$. E_{A1}/R = 250 K; E_{A2}/R = 150 K and E_{A3}/R = 200 K. **A** enters the system at T_0 = 300 K at a molar flow rate of 20 mol/s and a concentration of 40 mol/m^3.

(a.) If the CSTR volume is 0.5 m^3 and the exit temperature is 373 K, find the concentrations of A and B (C_A and C_B).

If, ΔH_1^0 (T_R) = -5 KJ/mol; ΔH_2^0 (T_R) = -5 KJ/mol; ΔH_3^0 (T_R) = -2.5 KJ/mol. T_R = 300 K. $C_{PA} = C_{PB} = C_{PC} = C_{PD}$ = 120 J/mol/K.

(b.) Find the expression you would need to solve to find the exit Temperature for the CSTR. Express your equation in terms of f(T) = 0. Assume CSTR volume = 0.5 m^3.

(c.) Find a new f(T) with ΔH_1^0 (T_R) = -5 KJ/mol and ΔH_2^0 (T_R) = ΔH_3^0 (T_R) = 0 KJ/mol. Find the approximate range in which the exit temperature T lies. This you can do using f(T). Change the value of T and find when the sign of f(T) changes from either positive value to negative or from negative value to positive. When f(T) changes sign it crosses through zero (f(T)=0). This will give you the range in which T is. I would suggest find f(290), f(300), f(310) and f(320) and see when the sign changes.

Homework # 7

1. (50 Points) The following reaction is occurring on the surface of a catalyst.

$$2A_{(g)} \rightarrow B_{(g)}$$

The equation below is found to represents the overall rate of reaction:

$$-r_A = \frac{K.P_A{}^2}{\left(1 + K_A P_A + K_B P_B\right)^2}$$

Suggest a reaction mechanism that would follow this equation and show how you would derive this equation using that mechanism.

2. (50 points) The water-gas shift reaction

$$CO + H_2O \Leftrightarrow CO_2 + H_2$$

has been hypothesized to occur by a Eley-Rideal mechanism involving oxidation and reduction of the catalyst surface:

$$H_2O + S \Leftrightarrow H_2 + OS$$
$$CO + OS \Leftrightarrow CO_2 + S$$

If CO oxidation is rate limiting, find an expression for the reaction rate in terms of gas phase concentrations.

Homework # 8

1. The Table below shows the various Fischer-Tropsch mechanism that have been proposed for conversion of carbon monoxide and hydrogen into straight chain polymers via nucleation of CH_2^*.

Primary elementary reactions scheme for FT synthesis.

Model	No.	Elementary reaction	Model	No.	Elementary reaction
FT-I	1	$CO + * \rightleftharpoons CO^*$	FT-X	1	$CO + * \rightleftharpoons CO^*$
	2	$CO^* + H_2 \rightleftharpoons CHOH^*$		2	$CO^* + * \rightleftharpoons C^* + O^*$
	3	$CHOH^* + H_2 \rightleftharpoons CH_2^* + H_2O$		3	$C^* + O^* + 2H_2 \rightleftharpoons CH_2^* + H_2O + *$
FT-II	1	$H_2 + * \rightleftharpoons H_2^*$	FT-XI	1	$CO + * \rightleftharpoons CO^*$
	2	$CO + H_2^* \rightleftharpoons CH_2^*O$		2	$CO^* + * \rightleftharpoons C^* + O^*$
	3	$CH_2^*O + * \rightleftharpoons CH_2^* + O^*$		3	$C^* + O^* + H_2 \rightleftharpoons CH_2^* + O^*$
	4	$O^* + H_2^* \rightleftharpoons H_2O + 2*$		4	$O^* + H_2 \rightleftharpoons H_2O + *$
				5	$O^* + CO^* \rightleftharpoons CO_2 + 2*$
FT-III	1	$H_2 + * \rightleftharpoons H_2^*$	FT-XII	1	$CO + 2* \rightleftharpoons C^* + O^*$
	2	$CO + H_2^* \rightleftharpoons CHOH^*$		2	$H_2 + 2* \rightleftharpoons 2H^*$
	3	$CHOH^* + H^* \rightleftharpoons CH_2^* + OH^*$		3	$C^* + O^* + 4H^* \rightleftharpoons CH_2^* + H_2O + 5*$
	4	$OH^* + H^* \rightleftharpoons H_2O + 2*$			
FT-IV	1	$CO + * \rightleftharpoons CO^*$	FT-XIII	1	$CO + 2* \rightleftharpoons C^* + O^*$
	2	$H_2 + 2* \rightleftharpoons 2H^*$		2	$H_2 + 2* \rightleftharpoons 2H^*$
	3	$CO^* + 3H^* \rightleftharpoons CH_2^*OH + 3*$		3	$C^* + O^* + 2H^* \rightleftharpoons CH_2^*O + 3*$
	4	$CH_2^*OH + * \rightleftharpoons CH_2^* + HO^*$		4	$CH_2^*O + * \rightleftharpoons CH_2^* + O^*$
	5	$OH^* + H^* \rightleftharpoons H_2O + 2*$		5	$O^* + H^* \rightleftharpoons OH^* + *$
				6	$OH^* + H^* \rightleftharpoons H_2O + 2*$
FT-V	1	$H_2 + 2* \rightleftharpoons 2H^*$	FT-XIV	1	$CO + 2* \rightleftharpoons C^* + O^*$
	2	$CO + 2H^* \rightleftharpoons CHOH^* + *$		2	$H_2 + * \rightleftharpoons H_2^*$
	3	$CHOH^* + * \rightleftharpoons CH^* + OH^*$		3	$C^* + O^* + H_2^* \rightleftharpoons CH_2^* + O^* + ^*$
	4	$CH^* + OH^* + 2H^* \rightleftharpoons CH_2^* + H_2O + 3*$		4	$CH_2^* + O^* + 2H^* \rightleftharpoons CH_3^* + HO^* + 2*$
				5	$OH^* + H^* \rightleftharpoons H_2O + 2*$
FT-VI	1	$H_2 + 2* \rightleftharpoons 2H^*$	FT-XV	1	$CO + 2* \rightleftharpoons C^* + O^*$
	2	$CO + H^* \rightleftharpoons COH^*$		2	$C^* + H_2 \rightleftharpoons CH_2^*$
	3	$COH^* + 2H^* \rightleftharpoons CH_2^*OH + 2*$		3	$O^* + H_2 \rightleftharpoons H_2O + *$
	4	$CH_2OH^* + H^* \rightleftharpoons CH_2^* + H_2O + *$			
	1	$H_2 + 2* \rightleftharpoons 2H^*$			
FT-VII	1	$CO + * \rightleftharpoons CO^*$	FT-XVI	1	$CO + * \rightleftharpoons CO^*$
	2	$H_2 + 2* \rightleftharpoons 2H^*$		2	$CO^* + * \rightleftharpoons C^* + O^*$
	3	$CO^* + H^* \rightleftharpoons CHO^* + *$		3	$H_2 + 2* \rightleftharpoons 2H^*$
	4	$CHO^* + H^* \rightleftharpoons CHOH^* + *$		4	$C^* + H^* \rightleftharpoons CH^* + *$
	5	$CHOH^* + H^* \rightleftharpoons CH_2^*OH + *$		5	$CH^* + H^* \rightleftharpoons CH_2^* + *$
	6	$CH_2^*OH + * \rightleftharpoons CH_2^* + HO^*$		6	$O* + H* \rightleftharpoons OH* + *$
	7	$OH^* + H^* \rightleftharpoons H_2O + 2*$		7	$OH^* + H^* \rightleftharpoons H_2O + 2*$
				8	$O^* + CO^* \rightleftharpoons CO_2 + 2*$
FT-VIII	1	$CO + * \rightleftharpoons CO^*$	FT-XVII	1	$CO + 2* \rightleftharpoons C^* + O^*$
	2	$H_2 + 2* \rightleftharpoons 2H^*$		2	$H_2 + * \rightleftharpoons H_2^*$
	3	$CO^* + 2H^* \rightleftharpoons HCOH^* + 2*$		3	$C^* + H_2^* \rightleftharpoons CH_2^* + *$
	4	$HCOH^* + * \rightleftharpoons CH^* + OH^*$		4	$O^* + H_2^* \rightleftharpoons H_2O + 2*$
	5	$CH^* + H^* \rightleftharpoons CH_2^* + *$			
	6	$OH^* + H^* \rightleftharpoons H_2O + 2*$			
FT-IX	1	$CO + * \rightleftharpoons CO^*$	FT-XVIII	1	$CO + 2* \rightleftharpoons C^* + O^*$
	2	$H_2 + * \rightleftharpoons H_2^*$		2	$H_2 + 2* \rightleftharpoons 2H^*$
	3	$CO^* + H_2^* \rightleftharpoons CH_2O^* + *$		3	$C^* + 2H^* \rightleftharpoons CH_2^* + *$
	4	$CH_2O^* + * \rightleftharpoons CH_2^* + O^*$		4	$O^* + 2H^* \rightleftharpoons CH_2^* + *$
	5	$O^* + H_2^* \rightleftharpoons H_2O + *$			
	6	$O^* + CO^* \rightleftharpoons CO_2 + 2*$			

Derive the rate expression for FT-VIII, if the third reaction (in FT-VIII) is the rate limiting step.

2. The following reaction is occurring on the surface of a catalyst.

$$2A_{(g)} + D_{(g)} \rightarrow B_{(g)} + C_{(g)}$$

The equation below is found to represents the overall rate of reaction:

$$-r_A = \frac{K.P_D P_A{}^2}{(1 + K_A P_A + K_B P_B)^2}$$

Suggest a reaction mechanism that would follow this equation and show how you would derive this equation using that mechanism.

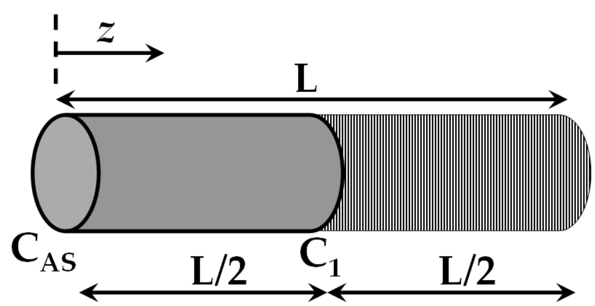
Homework # 9

1. (50 points) The reaction rate of a 1^{st}-order, elementary catalytic reaction ($A \rightarrow B$) was measured in a well-stirred reactor in which 5 grams of 0.1 mm catalyst particles were suspended. For a concentration of reactant of 1×10^{-3} mol/L, the reaction rate was found to be 1×10^{-4} mol/g/s.

   a) When the stirring rate was increased, the reaction rate did not change. What does it say about the reaction?
   b) Show that these data were obtained without the effects of pore-diffusion limitation.
   c) This reaction is to be run in a packed-bed reactor that is 10 m long and contains 10 cm diameter tubes. What particle diameter should be used in this reactor to get 75% conversion? Will there be pore-diffusion limitation? Assume that there is no pressure drop.

   Useful information:
   $D_e = 10^{-1}$ cm^2/s (De is constant no matter what size particles are used)
   ρ_B = (bulk density of catalyst in the packed bed) = 1 g/cm^3 (same for any particle size)
   ρ_C (catalyst density) = 2 g/cm^3 (same for any particle size)
   $C_{Ao} = 1 \times 10^{-3}$ mol/L
   F_{Ao} = 1 mol/s

2. (50 points) A pore of diameter d and length L on catalyst surface has degraded over time and is now half poisoned. Due to poisoning, the first half of the pore (till z = L/2) is non-reactive. The concentration at the inlet is C_{AS} and at the L/2 is C_1 (known). The diffusivity of the reactant is D_e. If the reaction is a first order reaction on the surface with a rate constant = k" (length/time units) then:



   a. Solve for concentration profile from z = 0 to L/2. State all the assumptions.
   b. If the effectiveness factor for no poisoning is $\eta_1 = \tanh(\phi_1)/\phi_1$, where $\phi_1 = L\sqrt{\dfrac{4k''}{dD_e}}$;

   calculate the new effectiveness factor for poisoning in terms of C_{AS}, C_1 and ϕ_1.

Homework # 10

1. (50 points) The activation energy of an isomerization reaction which obeys 1^{st}-order kinetics is known to be 100 kJ/mol and the pre-exponential of reaction is 1×10^5 cm/min. When the

isomerization reaction is run under strong pore diffusion limitations at 400 K, 500 g of catalyst are required to achieve 50% conversion.

a) Write the mole balance for the packed-bed reactor.
b) Calculate how much catalyst is required to get 50% conversion if the temperature is raised to 425 K

You should assume that there are no external mass transfer limitations, but that strong pore diffusion limitations are present at 425 K. The catalyst density (r_c) is 2.5 g/cm^3 and the bed void fraction (e) is 0.4.

2. (50 points) The irreversible gas-phase reaction

$$A \leftrightarrow B$$

is carried out over a packed bed of solid catalyst particles. The reaction is first order in the concentration of A on the catalyst surface:

$$-r_{As}' = k'C_{As}$$

The feed consists of 50% (mole) A and 50% inerts and enters the bed at a temperature of 300 K. The entering volumetric flow rate is 10 dm^3/s (i.e. 10,000 cm^3/s). The relationship between the Sherwood number and the Reynolds number is:

$$Sh = 100\ Re^{1/2}$$

As a first approximation, one may neglect pressure drop. The entering concentration of A is 1.0 M. Calculate the catalyst weight necessary to achieve 60% conversion of A for isothermal operation.

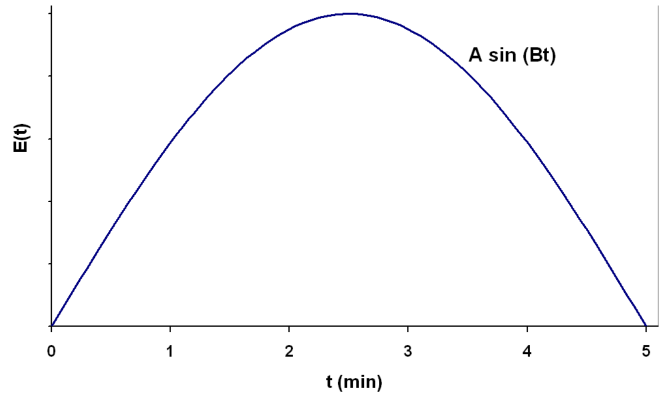
Additional Information:
Kinematic viscosity: $\nu = 0.02$ cm^2/s
Particle diameter: $d_p = 0.15$ cm
Superficial velocity: U = 9 cm/s
Catalyst surface area/mass of catalyst bed: a = 60 cm^2/g
Diffusivity of A: $D_e = 10^{-2}$ cm^2/s
k' (300 K) = 0.01 cm^3/s/g

Homework # 11

3. (50 points) A first order reaction

$$A \xrightarrow{k} B$$

with k = 2 min^{-1} is taking place in a real reactor with the following RTD function



For $0 < t < 5$ min; $E(t) = A \sin(Bt)$
$t > 5$ min; $E(t) = 0$

(a) What is the mean residence time?
(b) What is the conversion predicted by the segregation model?
(c) If there is no reaction taking place, what is the probability that a molecule entering the reactor spends a time between 5/3 min and 5/2 min inside the reactor?

4. (40 points) An existing stirred-tank reactor is to be used for a 2nd-order liquid phase reaction. A tracer injection led to the following data:

t	0	60	120	180	240	300	360	420	480	540	600	720	840
C	0	1	5	8	10	8	6	4	3	2.2	1.5	0.6	0

where C is concentration in moles/L and t is the time in seconds.

If the reaction has been found to give 80% conversion in 400 seconds in a batch reactor with C_{Ao}=0.5 mol/L, estimate the conversion that can be achieved in the stirred-tank reactor that gave the tracer data shown above using the complete segregation model. Assume the same inlet concentration as in the batch reactor.

PROJECT:

OBJECTIVE:

Design a reactor to produce Maleic-anhydride from benzene.

REACTION DETAILS:

Maleic anhydride is an industrially important commodity chemical used primarily in the production of polyester resins. An old commercial process to produce maleic anhydride employs a vanadium-phosphorous-oxide catalyst (VPO) to selectively oxidize butane:

$$C_6H_6 + 4.5\ O_2 \rightarrow C_4H_2O_3 + 2\ CO_2 + 2\ H_2O \quad \Delta H_R = -1236\ \text{kJ/mol} \tag{1}$$

However, side reactions to produce CO and CO_2 are also important:

$$C_6H_6 + 7.5\ O_2 \rightarrow 6\ CO_2 + 3\ H_2O \quad \Delta H_R = -2656\ \text{kJ/mol} \tag{2}$$

$$C_4H_2O_3 + 2\ O_2 \rightarrow 2\ CO + 2\ CO_2 + H_2O \quad \Delta H_R = -2656\ \text{kJ/mol} \tag{3}$$

REACTION KINETICS

Reaction rates for these three reactions are given by:

$$r_1 = k_1 P_1^{\alpha 1}/(1+K_2P_2^2)$$
$$r_2 = k_2 P_1^{\alpha 2}$$
$$r_3 = k_3 P_2/(1+K_2P_2)^2$$

where, P_1 is the partial pressure of benzene and P_2 is the partial pressure of maleic anhydride.

The following kinetic parameters are suggested (rate constants at 400°C):

$k_1 = 1 \times 10^{-6}$ kmol/kg s atm$^{0.35}$
$k_2 = 1.5 \times 10^{-7}$ kmol/kg/s atm$^{0.35}$
$k_3 = 2.9 \times 10^{-6}$ kmol/kg s atm
$E_1 = E_2 = 83100$ J/mol
$E_3 = 155000$ J/mol
$\alpha_1 = \alpha_2 = 0.35$
$K_2 = 310$ atm^{-1}

REACTOR CONFIGURATION

The reactor to be designed is a packed bed reactor with multiple reaction tubes with external cooling. External co-current cooling is to be provided by a mixed salt flowing at 2000 Kg/h. The tube diameter is to be 25 mm OD to maximize heat transfer (Length = 12 ft and thickness =1/8”). Heat transfer is a significant issue in this reactor due to the large exothermicity of the oxidation reactions.

Cp (mixed salt) = $A + BT + CT^2 + DT^3$ (kJ/kmol/K)

	A	B	C	D	[MW]
Mixed Salt	50	$2x10^{-3}$	$1\ x10^{-5}$	$-4x10^{-9}$	28

The heat capacities for the other components can be found from the NIST website (http://webbook.nist.gov/) in Thermochemistry section (find the Shomate equations).

DESIGN BASIS

The feed (assume physical properties of air) will consist of 15 % benzene in a mixture of oxygen and nitrogen (1:1). Total inlet pressure is 2 atm, and the total molar flow rate is 0.1 kmol/hr (per tube). Pressure drop should be considered using the Ergun equation. The following information may be helpful for the design:

Overall heat transfer coefficient (based on outside area) = 120 W/m^2 K
Catalyst diameter = 2 mm
Void fraction (ε) = 0.45
ρ_B=3.2 x 10^3 kg/m^3
Changing the catalyst diameter by a factor of 10 did not affect reactor performance.

DELIVERABLES

You are to consider your design in several parts, and satisfy the following requirements:

1. Model the maleic anhydride reactor assuming an inlet reactant temperature of 300^{o}C and inlet salt temperature of 150^{o}C, calculating the conversion of benzene. Plot the pressure, reactor-temperature, salt-temperature and molar flow rates of all the components as functions of reactor length.

2. What will be the analysis for part 1 with counter-current heat exchanger?

3. Plot the following:
    a. The Pressure Profile for different particle sizes (1, 2, 3, and 4 mm)
    b. Inlet salt temperature (50, 100, 150, 200, 250, 300 and 350 ^{0}C) as a function of the reactor exit temperature and final conversion (for 2 mm particle size).

4. As an engineer, what changes to the design and operating parameters will you suggest to increase the conversion to 85-95%?

5. What safety concerns are there for this reaction? How could you design your reactor to account for these concerns? For example, you might consider the effects of varying the salt inlet temperature on the resulting temperature profiles.

The final product of your work should be a detailed, professional report showing your results and analysis. In this report, you should specifically address the following: design basis, procedure used to solve the problem (including equations used and assumptions made), results, and conclusions/recommendations. Detailed calculations necessary to show how the problem was solved should be included at the end of the report in a sample calculation section.

Note that your grade for this project will be based on the accuracy of your results and completeness of your report. Also note that this is to be a detailed design, so while some assumptions will be acceptable. However, some assumptions that we have made for other projects (constant density, constant heat capacity, no pressure drop) should not be made for your final design.